KKT / competitor time — every comparator and every case

Reference KKT · elapsed

	Compacting Borůvka	Kruskal	Prim	Original Borůvka
Leaf hierarchy | n=65,536, d=16	1.18	1.51	1.83	1.54
Leaf hierarchy | n=1,048,576, d=16	0.83	3.77	3.02	1.89
Leaf distance | n=65,536, d=16	1.18	1.48	1.33	1.59
Leaf distance | n=1,048,576, d=16	0.83	3.72	2.19	1.92
Shuffled leaf core | n=262,144, d=16	1.12	2.21	2.33	3.23
Random leaf weights | n=262,144, d=16	1.11	1.61	1.47	2.33
Bundled hierarchy | n=262,144, d=64	1.01	1.79	1.84	1.89
Random bundled weights | n=262,144, d=64	1.04	2.78	2.92	6.22
Connected G(n,m) | n=65,536, d=128	1.05	1.82	3.24	5.21
Connected G(n,m) | n=262,144, d=16	1.13	2.01	0.97	2.47

Candidate KKT · elapsed

	Compacting Borůvka	Kruskal	Prim	Original Borůvka
Leaf hierarchy | n=65,536, d=16	1.05	1.35	1.64	1.36
Leaf hierarchy | n=1,048,576, d=16	0.40	1.76	1.45	0.90
Leaf distance | n=65,536, d=16	1.06	1.35	1.22	1.45
Leaf distance | n=1,048,576, d=16	0.40	1.77	1.06	0.89
Shuffled leaf core | n=262,144, d=16	0.66	1.27	1.37	1.87
Random leaf weights | n=262,144, d=16	0.70	1.04	0.95	1.54
Bundled hierarchy | n=262,144, d=64	0.87	1.54	1.59	1.64
Random bundled weights | n=262,144, d=64	0.49	1.27	1.35	2.86
Connected G(n,m) | n=65,536, d=128	0.54	1.03	1.85	2.99
Connected G(n,m) | n=262,144, d=16	0.70	1.23	0.60	1.52

Reference KKT · CPU

	Compacting Borůvka	Kruskal	Prim	Original Borůvka
Leaf hierarchy | n=65,536, d=16	1.18	1.51	1.83	1.54
Leaf hierarchy | n=1,048,576, d=16	0.83	3.77	3.02	1.89
Leaf distance | n=65,536, d=16	1.18	1.48	1.33	1.59
Leaf distance | n=1,048,576, d=16	0.83	3.72	2.19	1.92
Shuffled leaf core | n=262,144, d=16	1.12	2.21	2.33	3.23
Random leaf weights | n=262,144, d=16	1.11	1.61	1.47	2.33
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Connected G(n,m) | n=65,536, d=128	1.05	1.82	3.24	5.21
Connected G(n,m) | n=262,144, d=16	1.13	2.01	0.97	2.47

Candidate KKT · CPU

	Compacting Borůvka	Kruskal	Prim	Original Borůvka
Leaf hierarchy | n=65,536, d=16	1.05	1.35	1.64	1.36
Leaf hierarchy | n=1,048,576, d=16	0.40	1.76	1.45	0.90
Leaf distance | n=65,536, d=16	1.06	1.35	1.22	1.45
Leaf distance | n=1,048,576, d=16	0.40	1.77	1.06	0.89
Shuffled leaf core | n=262,144, d=16	0.66	1.27	1.37	1.87
Random leaf weights | n=262,144, d=16	0.70	1.04	0.95	1.54
Bundled hierarchy | n=262,144, d=64	0.87	1.54	1.59	1.64
Random bundled weights | n=262,144, d=64	0.49	1.27	1.35	2.86
Connected G(n,m) | n=65,536, d=128	0.54	1.03	1.85	2.99
Connected G(n,m) | n=262,144, d=16	0.70	1.23	0.60	1.52

4×
2×
1×
0.5×
0.25×

Below 1 (blue) favors KKT; above 1 (red) favors the comparator. Median of each graph's paired-trial median.
Two graph seeds × five timed repetitions. Same color scale across builds and clocks; no losses omitted.